

Mohammad Uzair Shah

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EDUCATION

Ph.D., Energy Science and Engineering Fall 2022 – Expected 2026
Bredesen Center, University of Tennessee, Knoxville, TN, USA.
CGPA: 3.97/4.00, Percentage: 99 %

Master of Science in Energy Systems Engineering Fall 2020 – Summer 2022
National University of Science and Technology (NUST), Islamabad, Pakistan
CGPA: 3.85/4.00, Percentage: 98 %

Bachelor of Science in Mechanical Engineering Fall 2015 – March 2020
Pakistan Institute of Engineering and Applied Sciences (PIEAS), Islamabad, Pakistan
CGPA: 3.56/4.00, Percentage: 89 %

Exchange School (U.S. Department of State sponsored), Mechanical Engineering Fall 2018
University of Missouri, Columbia, MO, USA
CGPA: 3.63/4.00, Percentage: 91 %

WORK EXPERIENCE

Graduate Research Assistant, Biomass Conversion and Modeling Lab, UTK Fall 2022 - present

- Developed a Python-based decision support model for sustainable energy crop production and optimal biorefinery siting.
- Designed an extensible framework that integrates techno-economic, environmental, and socio-economic criteria to support bioenergy development.
- Leveraged quantitative tools to optimize socio-economic and demographic factors, maximizing community benefits and supporting underserved communities.
- Incorporated Input-Output economic impact assessments to enhance socio-economic benefits through energy crop and biorefinery siting.
- Addressed environmental and energy justice issues by analyzing decision support model outcomes to inform policy-making.

Research Projects

Integrated Decision Support System Modeling for Biorefinery Siting Analysis, Dissertation project

- Designing and developing an integrated python-based, one-stop decision support model capable of integrating crucial techno-economic, environmental, and social parameters for sustainable siting of energy crops and biorefineries to optimize the investors and community benefits along with uplifting the underserved communities.

- Utilizing the model results to recommend bio-energy policy amendments for enhancing energy and environmental justice

Reducing Agricultural Carbon Intensity and Protecting Algal Crops (RACIPAC), DOE funded project

- Investigated feedstock-process-biochar-soil interactions to enhance carbon sequestration and agronomic benefits in bioenergy crop systems across the Southeast
- Contributed to the assessment of biochar and poultry litter impacts on Miscanthus and biomass sorghum for sustainable aviation fuel production, using machine learning, techno-economic analysis, and life cycle assessment.

A techno-economic and socio-environmental planning of wind turbines for sustainable development in Pakistan, Master's thesis project

- Integrated traditional techno-economic and life-cycle analysis with the crucial quantitative and qualitative social factors for sustainable siting of wind farms while recommending policy suggestions to improve development and social acceptance of wind farms in the community

PUBLICATIONS

Peer-reviewed Journal Articles

- Shah, M. U., Khanum, S., Waqas, A., Janjua, A. K., & Shakir, S. (2023). A techno-economic and socio-environmental planning of wind farms for sustainable development and transition to a decarbonized scenario: Pakistan as a case study. *Sustainable Energy Technologies and Assessments*, 55, 102969, <https://doi.org/10.1016/j.seta.2022.102969>.

CONFERENCE PRESENTATIONS

- **“Integrated GIS-based decision support model for optimal biomass preprocessing site identification using sociodemographic, economic, and environmental analysis”** (Oral presentation)
American Society of Agricultural and Biological Engineers (ASABE) Annual International Meeting, Anaheim, CA, July 2024
- **“Identification of potential Miscanthus x giganteus (Mxg) production areas in Tennessee using an integrated GIS-based multicriteria decision support system modeling”** (Poster presentation)
S1075 Multi-state Project 2024 Annual Meeting, Brookings, SD, July 2024
- **“GIS-based multi-criteria decision support for energy crop system analysis”** (Poster presentation)
Frontiers in Biorefining, St. Simons Island, GA, September 2024

LEADERSHIP EXPERIENCE

Cultural ambassador, U.S. Department of State, Washington, DC Fall 2018

- Participated in cultural exchange program UGRAD (1% acceptance rate) sponsored by U.S. Department of State to portray positive image of both countries and share cultural aspects

President, PIEAS Thematic Society (PTS)

2017-2018

- Headed various inter- and intra-university events at PIEAS including Synergy'17, PION'18 and Megafest'17 for community outreach and development

SKILLS

Python, ArcGIS (geospatial analysis software), data curation and analysis, process modeling, renewable energy siting, socio-economic analysis, I-O modeling, software development, techno-economic analysis, life cycle assessment, survey design, community outreach, interviewing